Accurately Characterizing a Light Emitting Diode

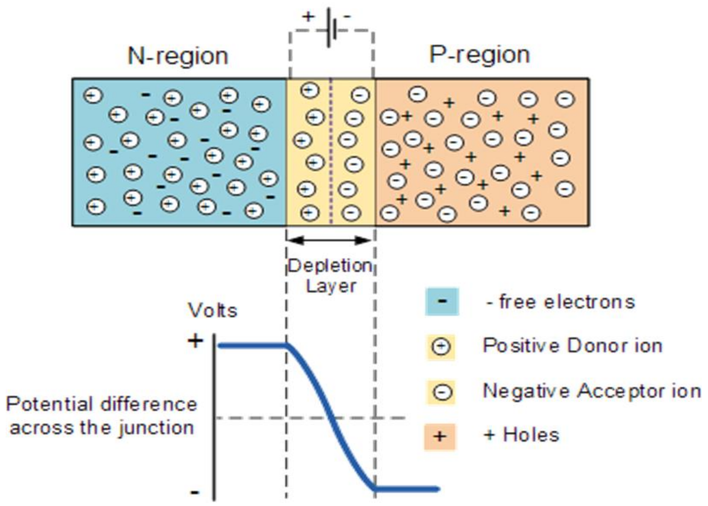
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Welcome to the Laser Center

Introduction

In the near future virtually all artificial light will come from light emitting diodes (LEDs), which have much higher energy efficiency and last far longer than current light sources, especially incandescent bulbs. LEDs are special semiconductor devices that produce light when a voltage is applied across them. The diode of an LED operates based on the same basic principles of a silicon diode.



The electrons in the semiconductor material are excited from an energy level called the valence band, where electrons are bound to atoms, to the conduction band, where electrons are free to move. The energy difference between the two bands is called the band gap energy(Eg). The electrons in an LED emit photons with energy Eg as they fall from the conduction band to the valence band in a process called radiative recombination. The band gap energy is determined by the materials the semiconductor is made of and in turn determines the wavelength of light produced. Already white LEDs light many of Stony Brook University's parking lots. The rapidly improving LED technology is destined to become the number one man-made light source within our lifetimes.

Exponential Current

Diodes only allow current to flow in one direction by applying positive voltage to the p-type and a negative voltage to the n-type. This forward biased voltage brings electrons with enough energy from the valence band to the conduction band. William Shockley was the first to derive the formula for the current flow

$$I = I_s\left[e^{\frac{eV}{nk_BT}} - 1\right]$$

Where I_s is the reverse saturation current, e is the charge of an electron, V is the voltage, n is an "ideality factor", k_B is Boltzmann's constant, and T is temperature. For forward biased voltages the -1 can be neglected as $\exp(eV/nk_BT) >> 1$.

According to the equation, the voltage through the diode increases exponentially with increasing voltage. This is due to the exponential distribution of the electrons. The probability of finding an electron at a given energy level, E, is given by the Fermi-Dirac equation:

$$P(E) = \frac{1}{e^{\frac{E-E_F}{k_BT}} + 1}$$

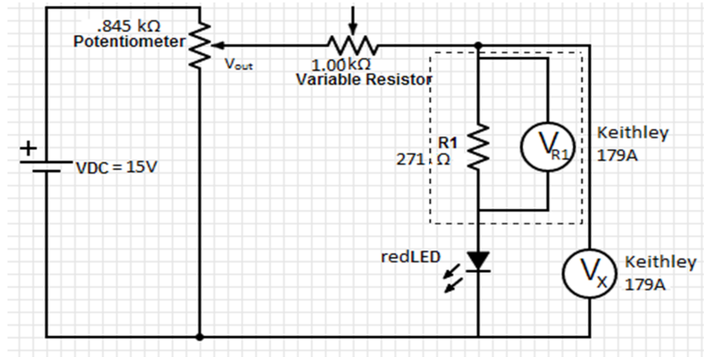
E_f is the Fermi energy

Applying a forward biased voltage to the diode decreases the potential barrier of the depletion zone allowing for electrons with less energy to cross it. According to the Fermi-Dirac equation the probability of finding an electron at a given E increases exponentially as E decreases. Therefore, the current will rise exponentially with increasing voltage.

Current vs Voltage Measurements

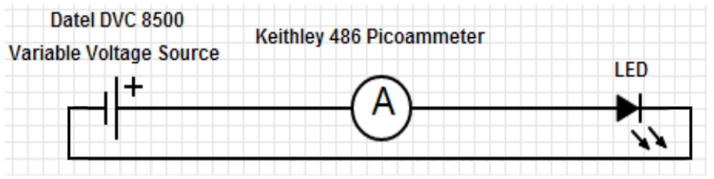
Two separate setups were used to measure the current and voltage of a red LED. The first circuit, taken from PHY-335, was used to measure currents above 10^{-6} amps through the LED. A second circuit was needed to measure nanoampere currents due to the limited sensitivity of the voltmeters and their finite internal resistance.

PHY 335 circuit

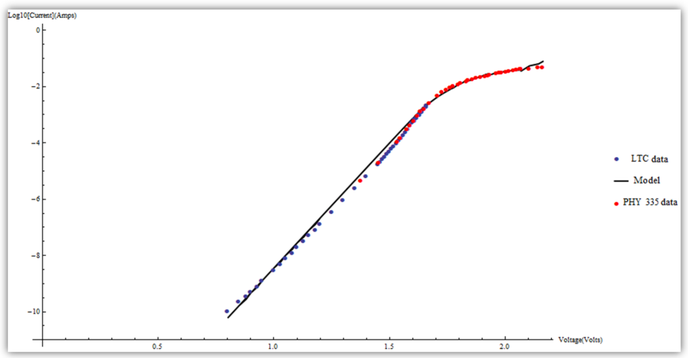


Using the circuit designed in the LTC, we were able to measure the current from 10^{-11} amps up to $2 * 10^{-3}$ amps, the max current rating of the picoammeter.

LTC Circuit



Results



At higher currents the temperature of the LED began to rise. We were able to successfully account for the rise in temperature in our model of the current. The is model is based on the log of Shockley's equation under the approximation $\exp(eV/nk_BT) >> 1$. The rise in temperature was found to be proportional to the power dissipation. The predominant form of heat dissipation in the LED is conductive cooling through the LED leads.

$$Log_{10}[I] = Log_{10}\left[I_s e^{\frac{eV}{nk_B(T+\alpha P)}}\right]$$

α is the constant for conductive cooling, T is the ambient temperature, and P is the power being dissipated by the LED.

Experimental Measurement of Planck's Constant

Planck' Constant, h = 6.626 * 10^{-34}Js, can be experimentally measured from the energy of the photons emitted from multiple LEDs. Measuring the voltage threshold of several different colored LEDs and plotting VTH vs c/ eλ will yield a line with slope h.. The voltage threshold for four LEDs were measured using the same LTC circuit for the I-V measurements.

$$E = \frac{hc}{\lambda}$$

$$E_g = \frac{hc}{\lambda} = eV_{TH}$$

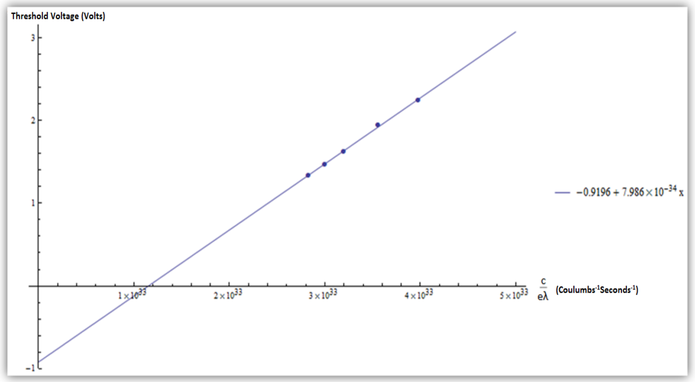
$$V_{TH} = h\frac{c}{e\lambda}$$

V_{TH} is the threshold voltage of the diode, c is the speed of light, e is the charge of an electron, and λ is the wavelength of the light emitted by a LED.

There is some ambiguity as to what the threshold voltage of an LED is. It is commonly referred to as the point at which the current starts to rise exponentially. As seen in the data fro m the I-V measurements, the current of an LED is always exponential for a forward biased voltage at constant temperature. The point at which the diode begins to emit light is another arbitrary choice of the voltage threshold. The intensity of the light emitted by the LED is proportional to the current and does not suddenly turn on. We chose to measure the voltage when we could see the LED first emit light.

Results

Wavelength	Voltage Threshold	Current	Power
nanometers	Volts	Amperes	Watts
585(Yellow)	1.63	4.20E-05	6.85E-05
660(Red)	1.34	5.93E-07	7.95E-07
624(Red)	1.48	1.63E-07	2.41E-07
525(Green)	1.95	2.14E-07	4.17E-07
470(Blue)	2.25	7.40E-08	1.67E-07



Planck's constant was found to be 7.99 E-34 Js from the slope. This result is accurate within 20.6% of the accepted value.

References

- Herrmann, Friedrich, Dr., and Peter Würfel, Dr. "Measuring Planck's Constant by Means of LED's." *KPK- Startseite*. N.p., n.d. Web. 25 Apr. 2014.
- Klipstein, Donald L. "Don's LED Main Page." *Don's LED Main Page*. N.p., 16 Feb. 2014. Web. 25 Apr. 2014.
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